

Wenhan (Danny) Guo

Baltimore, MD (+1) 626-417-8239 wguo28@jh.edu

EDUCATION

Johns Hopkins University , PhD student <i>Electrical and Computer Engineering</i>	Baltimore, MD 2025 – Present
Pomona College <i>B.A. in Physics (cum laude)</i>	Claremont, CA 2021 – 2025

PUBLICATIONS

Papers

- W. Guo**, Y. Gao, & Y. Sun, “ScoreField: Neural Inverse Scattering with Score-Based Generative Priors,” *ArXiv.org*.
<https://arxiv.org/abs/2608.02937>.
- W. Guo**, J. Yu, Y. Wang, J. U. Kang, and Y. Sun, “PSI3D: Plug-and-Play 3D Stochastic Inference with Slice-Wise Latent Diffusion Prior,” *2025 59th Asilomar Conference on Signals, Systems, and Computers*, 484-488.
<https://doi.org/10.1109/IEEECONF67917.2025.11443921>.
- Y. He, **W. Guo**, D. Wei, E. Bell, J. Yi, and Y. Sun, “VOLT: Volumetric Wide-Field Microscopy via 3D-Native Probabilistic Transport,” *ArXiv.org*. <https://arxiv.org/abs/2604.18807>.
- Y. Gao, **W. Guo** and Y. Sun, “Neural Inverse Scattering with Score-Based Regularization,” *2025 IEEE Conference on Computational Imaging Using Synthetic Apertures (CISA)*, College Park, MD, USA, 2025, pp. 1-5.
<https://doi.org/10.1109/CISA64343.2025.11091557>.

Posters

- W. Guo**, M. Cui, L. V. Wang, “Enhancing Photoacoustic Tomography with Phase-Encoding Mask,” *SURF at Caltech*. **Vodopia-Hasson Best Poster Prize**.
- W. Guo**, N. Doan, P. Choi, P. Evans, N. Saini, C. Zhai, M. Shao, “Developing an Automated Observing System for the Table Mountain 1-m Telescope,” *Bulletin of the AAS*, 55(2). <https://baas.aas.org/pub/2023n2i177p56>.

RESEARCH EXPERIENCE

Johns Hopkins University , Hopkins Computational Imaging Group Graduate Research Assistant, Advisor: Yu Sun <i>Computational Imaging with Score-based Methods</i>	Baltimore, MD 2025 – Present
California Institute of Technology , Caltech Optical Imaging Laboratory Research Assistant, Advisor: Lihong V. Wang <i>Phase-Encoding Mask for Photoacoustic Computed Tomography</i>	Pasadena, CA Summer 2024
Pomona College & JPL , Table Mountain Observatory Research Assistant, Advisor: Philip Choi <i>Automation for 1-Meter-Grade Optical Telescopes</i>	Wrightwood, CA 2022 – 2023

TEACHING & OUTREACH

ASTR 051: Advanced Introductory Astronomy, TA	Fall 2022, Fall 2024
ASTR 101: Techniques in Observational Astrophysics, TA	Spring 2023
Manager & Telescope Trainer: Pomona College Brackett Observatories	2022 – 2025

MENTORSHIP

- Johns Hopkins University**
Kewei Zhang (M.S. BME, 2026-present)
Junxiu Lu (Visiting Student, 2026-present)
Yetao He (M.S. BME, 2025-2026)

AWARD & HONORS

Virginia and Edward Wysocki, Sr. Fellowship in ECE	2026
The John Stauffer Scholarship for Academic Merit (<i>Highest academic promise in math and science</i>)	2025
The Frank Parkhurst Brackett, Jr., and Davida Wark Brackett Prize (<i>Excellence in Astrophysics</i>)	2025
Vodopia-Hasson Prize (<i>Caltech best poster</i>)	2024
Caltech SURF Grant	2024
The F.S. Jennings Memorial Prize in Critical Writing (x2) (<i>Excellence in English literature</i>)	2022, 2023
Pomona College SURP Grant	2022
The Moncrieff Astronomy Prize	2022

SKILLS

Technical Skills: PyTorch, MATLAB, Mathematica, Tensorflow, CAD design
Language: English, Chinese